Basic Relations

Position: $\mathbf{r}(t)$

Displacement {from A to A'}:

$$\Delta\mathbf{r} = (\mathbf{r} + \Delta\mathbf{r}) - \mathbf{r}$$

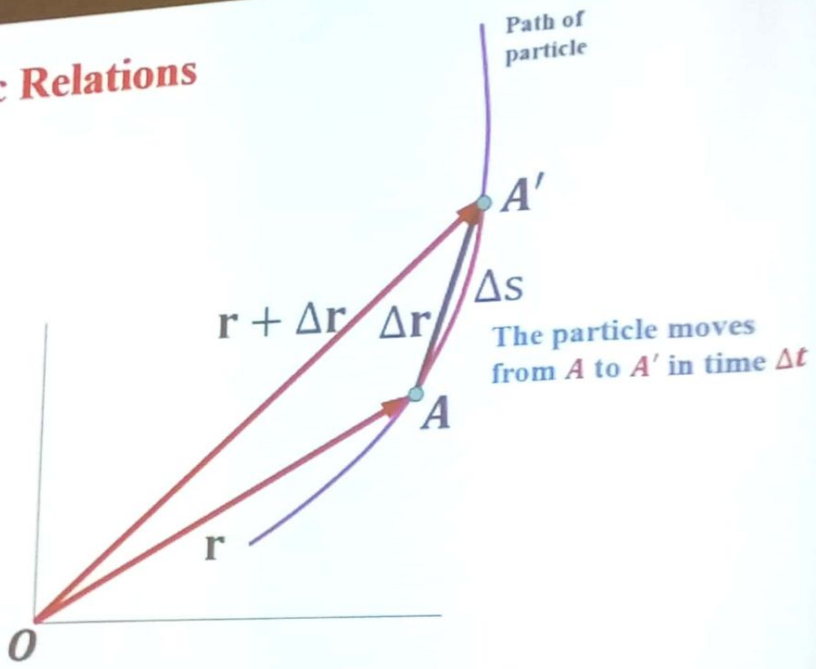
Distance traveled: Δs

Average Velocity:

$$V_{av} = \frac{\Delta\mathbf{r}}{\Delta t}$$

Instantaneous Velocity:

$$V = \lim_{\Delta t} \frac{\Delta\mathbf{r}}{\Delta t} = \frac{d\mathbf{r}}{dt}$$



Basic Relations





$$\Delta \mathbf{V} = \mathbf{V}' - \mathbf{V}$$

Average Acceleration: $\boldsymbol{a_{av}} = \dfrac{\Delta \mathbf{V}}{\Delta t}$

Instantaneous Acceleration: $\boldsymbol{a} = \lim_{\Delta t} \dfrac{\Delta \mathbf{V}}{\Delta t} = \dfrac{d\boldsymbol{V}}{dt}$

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Rectangular Coordinates (x-y)

Position

$$\mathbf{r} = x\mathbf{i} + y\mathbf{j}$$

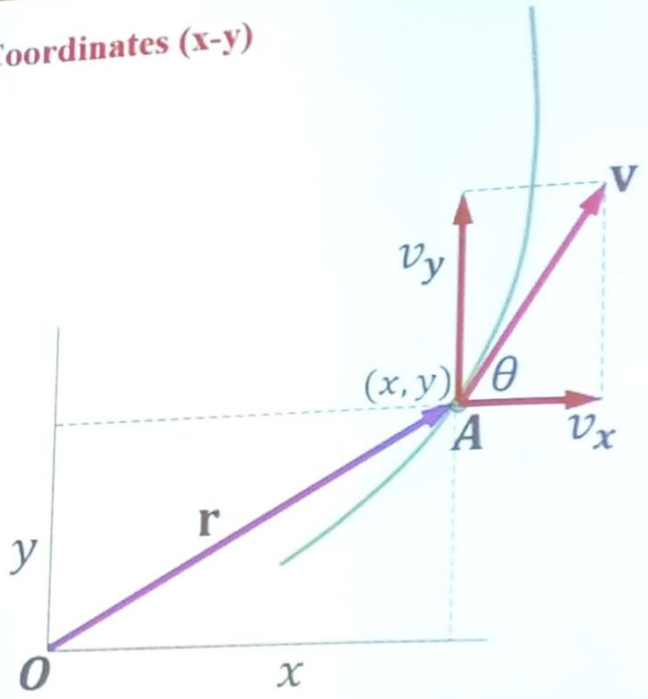
Velocity

$$\mathbf{V} = \frac{d\mathbf{r}}{dt} = \frac{dx}{dt}\mathbf{i} + \frac{dy}{dt}\mathbf{j}$$

$$= v_x\mathbf{i} + v_y\mathbf{j}$$

Speed

$$v = \sqrt{(v_x)^2 + (v_y)^2}$$



Acceleration

$$\mathbf{a} = \frac{dv_x}{dt}\mathbf{i} + \frac{dv_y}{dt}\mathbf{j}$$

$$= a_x\mathbf{i} + a_y\mathbf{j}$$

magnitude

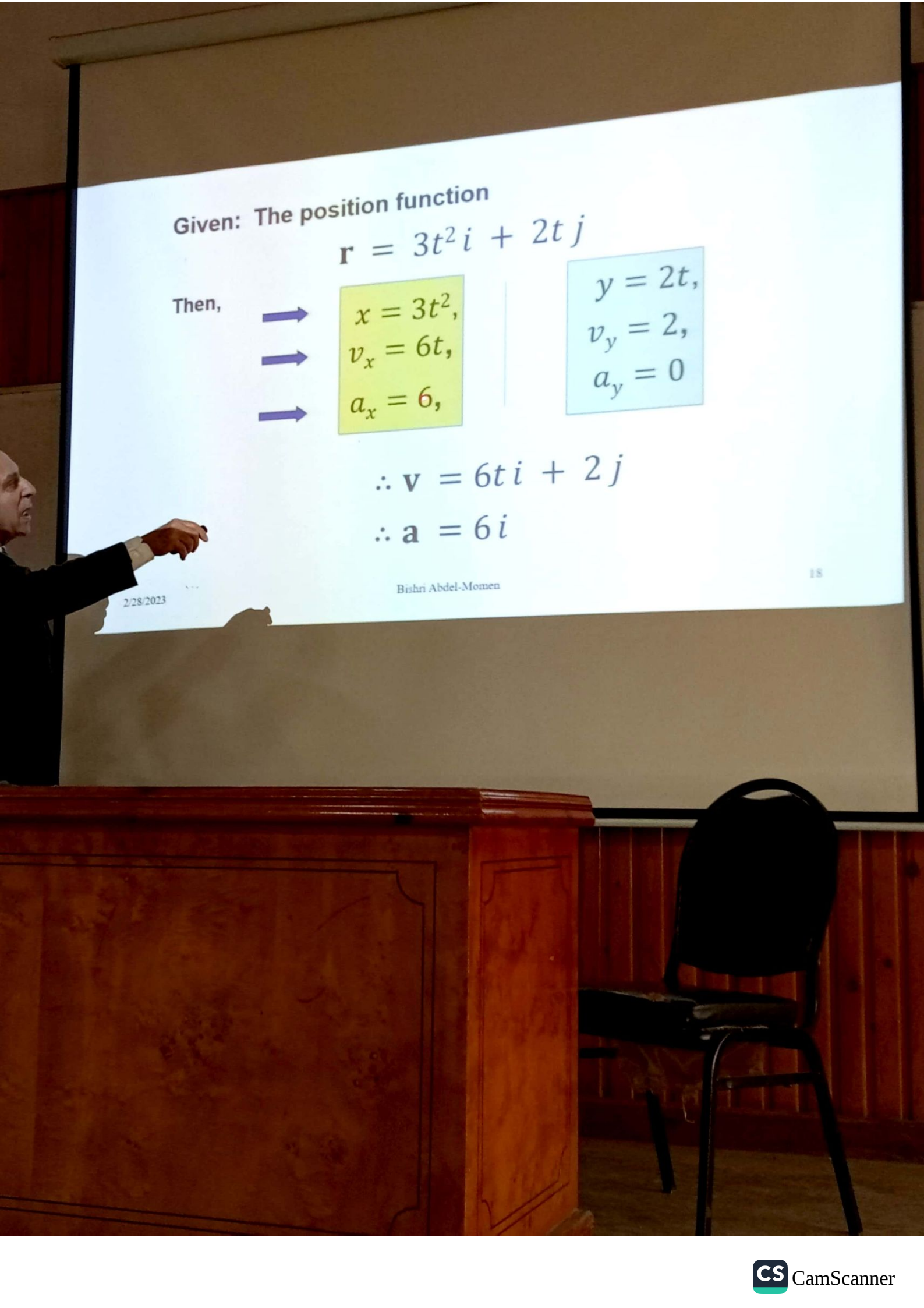
$$a = \sqrt{(a_x)^2 + (a_y)^2}$$

Direction

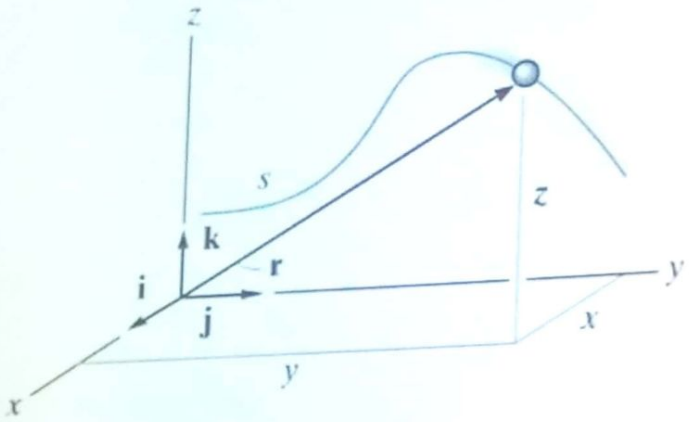
$$\tan\theta = \frac{a_y}{a_x}$$



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Given: The position function
$\mathbf{r} = 3t^2 i + 2t j$
Then,
$x = 3t^2,$
$v_x = 6t,$
$a_x = 6,$
$y = 2t,$
$v_y = 2,$
$a_y = 0$
$\therefore \mathbf{v} = 6t i + 2 j$
$\therefore \mathbf{a} = 6 i$
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Rectangular Coordinates (x-y-z)



Position

$$\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$$

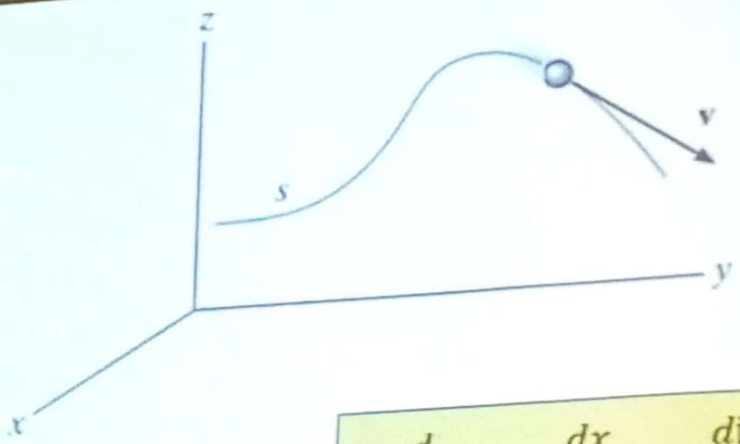
$$r = \sqrt{(x)^2 + (y)^2 + (z)^2}$$

Velocity

$$\mathbf{v} = \frac{d\mathbf{r}}{dt}$$

$$= \frac{d}{dt}(x\mathbf{i}) + \frac{d}{dt}(y\mathbf{j}) + \frac{d}{dt}(z\mathbf{k})$$

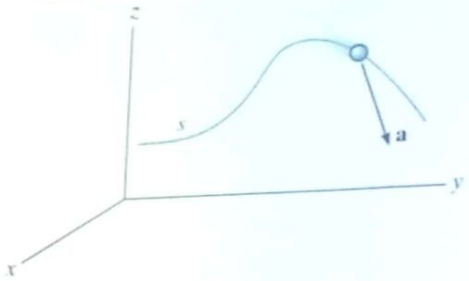
$$= v_x\mathbf{i} + v_y\mathbf{j} + v_z\mathbf{k}$$



$$\frac{d}{dt}(x\mathbf{i}) = \frac{dx}{dt}\mathbf{i} + x\frac{d\mathbf{i}}{dt}$$

$$= \frac{dx}{dt}\mathbf{i} + 0$$

Acceleration



$$\mathbf{a} = \frac{d\mathbf{V}}{dt}$$

$$= a_x\mathbf{i} + a_y\mathbf{j} + a_z\mathbf{k}$$

$$a_x = \frac{dv_x}{dt}, \quad a_y = \frac{dv_y}{dt}, \quad a_z = \frac{dv_z}{dt}$$

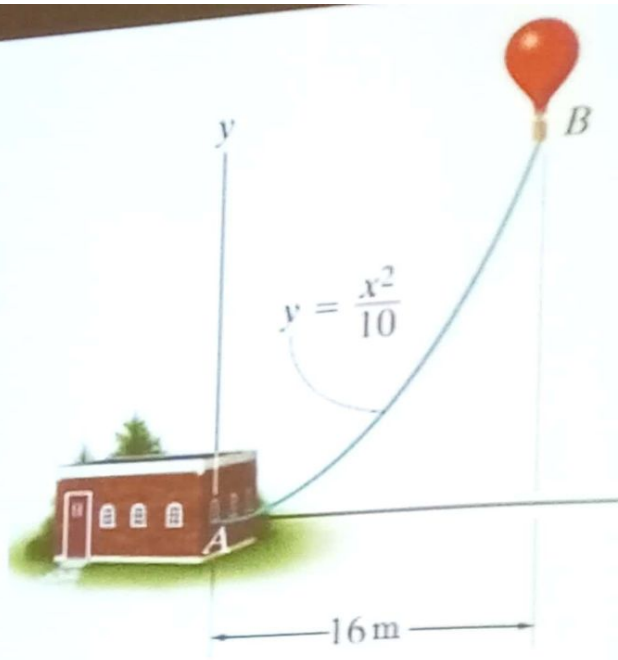
The acceleration magnitude

$$a = \sqrt{(a_x)^2 + (a_y)^2 + (a_z)^2}$$

The direction is specified by the unit vector $\mathbf{u}_a = \frac{\mathbf{a}}{a}$

Example

At any instant the horizontal position of the weather balloon is defined by $x = (8t)$ m, where t is in seconds. If the equation of the path is $y = x^2/10$, determine the magnitude and direction of the velocity and the acceleration when $t = 2$ s.



Velocity. The component in the x direction is

$$v_x = \frac{d}{dt}(x) = \frac{d}{dt}(8t) = 8 \quad \text{m/s}$$

$$v_y = \frac{d}{dt}(y) = \frac{d}{dt}(x^2/10) = \frac{d}{dx}(x^2/10)\frac{dx}{dt}$$

$$= \frac{2x}{10}v_x = \frac{16x}{10} = \frac{8(8t)}{5} = \frac{8(16)}{5} = 25.6 \text{ m/s}$$

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$$\therefore\ v = \sqrt{(8)^2 + (25.6)^2} = 26.8\ \text{m/s}$$ **Ans.**

The direction is tangent to the path, where

$$\theta_v = \tan^{-1}\left(\frac{v_y}{v_x}\right) = \tan^{-1}\left(\frac{26.8}{8}\right) = 72.6^0$$ **Ans.**

Acceleration.

$$a_x = \frac{d}{dt}(v_x) = \frac{d}{dt}(8) = 0$$

$$a_y = \frac{d}{dt}(v_y) = \frac{d}{dt}(2xv_x/10)$$

$$v_y = \frac{2x}{10}v_x$$

$$= \frac{1}{10}(2xa_x + 2v_xv_x)$$

$$= \frac{1}{10}(2 * (2 * 8) * 0 + 2(8)^2) = 12.8\ \text{m/s}^2$$

Example

For a short time, the path of the plane is described by $y = (0.001x^2)$ m. If the plane is rising with a constant velocity of 10 m/s, determine the magnitudes of the velocity and acceleration of the plane when it is at $y = 100$ m



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SOLUTION

When $y = 100$ **m, then**

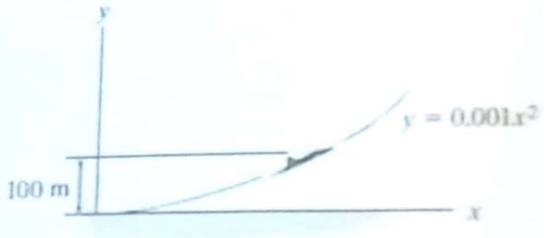
$$100 = 0.001x^2 \rightarrow \quad x = 316.2 \text{ m.}$$

Also, since $v_y = 10\ m/s$**, then**

$$y = v_y t \rightarrow; \quad 100\ m = (10\ m/s)\ t \rightarrow \quad t = 10\ s$$

locity.

$$v_y = \frac{d}{dt}(y) = \frac{d}{dt}(0.001x^2) = 0.002x\frac{dx}{dt} = 0.002x\ v_x$$



Thus

$$10 = 0.002(316.2)\ v_x$$

$$\therefore v_x = 15.81\ m/s$$

The magnitude of the velocity is therefore

$$v = \sqrt{(v_x)^2 + (v_y)^2} = \sqrt{(15.81)^2 + (10)^2} = 18.7\ \text{m/s}$$

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$$v_y = 0.002x\, v_x$$
$$v_x = 15.81\ m/s$$

Acceleration.

$$a_y = \frac{d}{dt}(v_y) = 0.002x\, a_x + 0.002 v_x v_x$$

$$= 0.002\left(x\, a_x + (v_x)^2\right)$$

When $x = 316.2$ m, $v_x = 15.81 \frac{m}{s}$, $a_y = 0$

$$\therefore\ 0 = 0.002\left(316.2\, a_x + (15.81)^2\right)$$

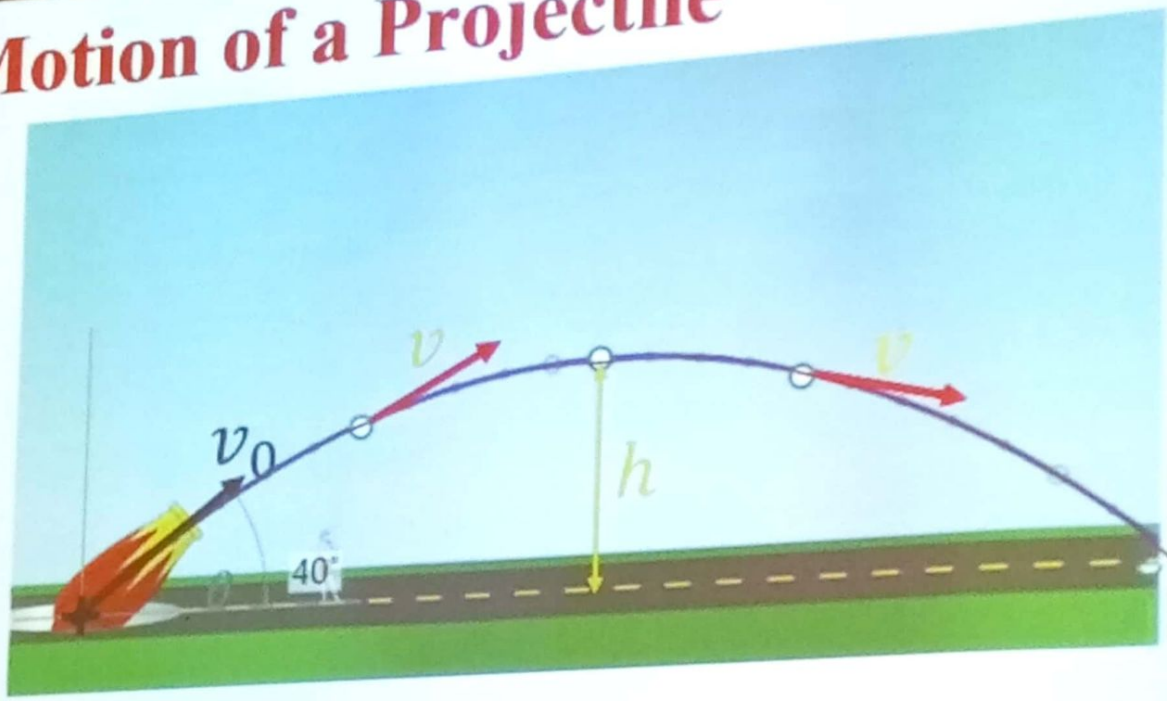
$$\therefore a_x = -0.791\ \ m/s^2$$

The magnitude of the plane's acceleration is

$$a = \sqrt{(a_x)^2 + (a_y)^2} = \sqrt{(-0.791)^2 + (0)^2} = 0.791\ \mathrm{m/s^2}$$

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Motion of a Projectile



Initial velocity: v_0

angel: θ

velocity: v

Max height: h

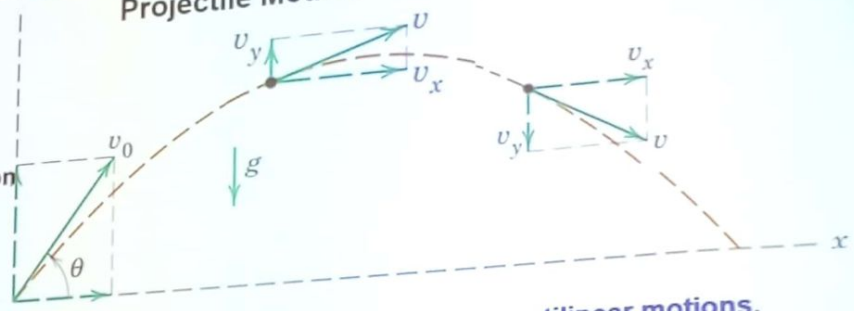
Range: R

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Projectile Motion

Assumptions

$a_x = 0$, $a_y = -g$ } constant acceleration



Projectile motion can be treated as the **superposition of two rectilinear motions.**

Horizontal Motion. ($a_x = 0$),